AI-POWERED INTELLIGENT TALENT ACQUISITION SYSTEM

INDIVIDUAL THESIS

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Individual component: Intelligent Resume Parsing & Skill Proficiency Analysis

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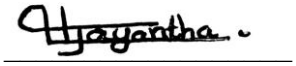
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ABSTRACT

Modern talent acquisition platforms still struggle to convert heterogeneous curriculum vitae (CV) documents into reliable, evidence-based skill profiles that recruiters and candidates can trust. Keyword-centric applicant tracking systems often reduce resumes to brittle term matches, while many commercial parsers extract fields without estimating skill proficiency or explaining why a score was assigned. This individual thesis presents the Intelligent Resume Parsing and Skill Proficiency Analysis component of an AI-Powered Intelligent Talent Acquisition System. The component implements a layered pipeline that extracts text from PDF, DOCX, and TXT resumes, parses structured sections, detects canonical skills against an alias-aware catalogue, and estimates proficiency using a fine-tuned multi-task DeBERTa-v3 model. The model jointly predicts project strength, skill-evidence strength, and experience–project alignment as continuous scores and as Beginner, Intermediate, and Advanced classes. Gemini-based explanations then ground the scores in CV evidence for human review. Experiments on a Sri Lankan localized resume skill-context dataset comprising 4,817 raw records expanded to 24,567 skill-context rows produced strong held out results: average regression MAE of 2.73 on a 0–100 scale, skill-strength classification accuracy of 98.98%, and average classification accuracy of 95.37% across the three task heads. The component is delivered as containerized microservices integrated with candidate profile, matching, interview, and attrition modules. Section 6.3 (User or stakeholder validation) is left blank pending completion of recruiter and candidate evaluation sessions. The individual contribution covers problem formulation, dataset preparation, model design and evaluation, parsing services, explainability integration, frontend skill-analysis experience, testing, and platform integration.

Keywords: Resume Parsing, Skill Proficiency Analysis, DeBERTa, Explainable AI, Talent Acquisition.

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LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
API	Application Programming Interface
CI/CD	Continuous Integration / Continuous Deployment
CV	Curriculum Vitae
DeBERTa	Decoding-enhanced BERT with Disentangled Attention
LLM	Large Language Model
JSON	JavaScript Object Notation
ATS	Applicant Tracking System
MAE	Mean Absolute Error
ML	Machine Learning
NLP	Natural Language Processing
OCR	Optical Character Recognition
QWK	Quadratic Weighted Kappa
REST	Representational State Transfer
RMSE	Root Mean Square Error
SaaS	Software as a Service
SDG	Sustainable Development Goal
UI	User Interface
XAI	Explainable Artificial Intelligence

1 INTRODUCTION

1.1 Overall project context

Talent acquisition is a multi-stage decision process that begins long before an interview is scheduled. Recruiters and hiring managers must interpret candidate documents, compare claimed competencies with role requirements, and decide who deserves scarce interview capacity. In practice, this process is still dominated by manual CV screening and keyword-based applicant tracking systems (ATS). Those tools are fast, but they frequently misrepresent candidate ability because they treat skill mentions as binary presence signals rather than evidence-graded proficiency statements [1], [2]. The group research project develops an AI-Powered Intelligent Talent Acquisition System that addresses recruitment as an integrated pipeline rather than as isolated analytics widgets. The complete solution, developed under Group ID R26 IT-082, combines resume intelligence, CV-to-job matching, interview analysis, and hiring attrition risk support. The shared product vision is to help organizations in Sri Lanka and similar labour markets make faster, fairer, and more transparent hiring decisions while retaining human authority over final offers. Within this platform, resume understanding is a foundational capability. Matching, interview preparation, and retention-risk modelling all depend on structured candidate representations that preserve skills, experience, projects, and confidence. If the resume layer is shallow, every downstream decision inherits noise. Consequently, the individual component reported in this thesis is positioned as the evidence-producing front end of the talent acquisition stack.

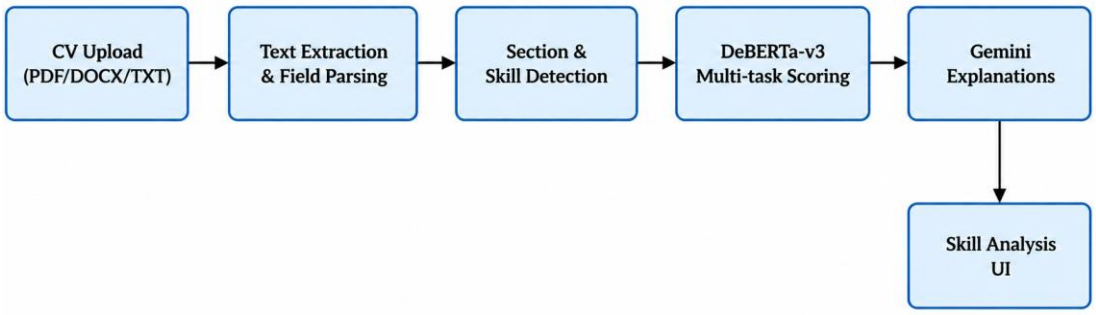


Figure 1:Intelligent resume parsing and skill proficiency analysis pipeline

1.2 Individual component context

The individual research component is Intelligent Resume Parsing & Skill Proficiency Analysis. It converts an uploaded resume into a structured, explainable skill profile. The pipeline extracts selectable text from PDF, DOCX, and TXT files; identifies candidate identity and sectioned content; detects technical and soft skills using a canonical catalogue with aliases; estimates proficiency through a multi-task transformer model; and returns Gemini-grounded explanations that cite CV evidence. The component is deliberately designed as a set of microservices rather than a monolithic script. Text extraction, profile analysis orchestration, strength scoring, and explanation generation are separated so that each concern can scale, fail independently, and integrate with the shared frontend. Candidate-facing pages expose an evidence-based skill analysis view, while recruiter and admin views consume the same structured outputs when reviewing applications. From a research perspective, the component investigates whether a localized, multi-task language model can estimate skill proficiency more usefully than rule only heuristics, while remaining operationally deployable inside a web platform. From a product perspective, it produces machine-readable outputs that other components can consume without re-parsing the original document.

1.3 Problem statement

How can a talent acquisition platform transform heterogeneous resume documents into structured, job-aware, and explainable skill proficiency profiles that remain robust under incomplete evidence, inconsistent CV formatting, and the need for human-verifiable justifications, without reducing candidate evaluation to opaque keyword matching? The problem is both technical and socio-technical. Technically, resumes vary in layout, language style, and evidence density. Socio-technically, candidates and recruiters need to understand why a skill was scored as Beginner, Intermediate, or Advanced. A system that is accurate but opaque is unlikely to be trusted in hiring workflows where fairness, contestability, and auditability matter [3], [4].

1.4 Research gap

Existing work and products leave four gaps that motivate this component. First, many ATS and parser products emphasize entity extraction or keyword hits, not proficiency estimation grounded in project and experience evidence [5], [6]. Second, generic NLP classifiers often score skills from isolated sentences and ignore cross section consistency between employment history and projects. Third, explainability is frequently absent or post-hoc, leaving end users with a score but no CV-linked rationale [7]. Fourth, Sri Lankan and regional hiring contexts need localized evidence and operational integration into an end-to-end acquisition platform rather than standalone notebook demos. This thesis therefore targets the gap between resume parsing and trustworthy skill proficiency analytics. The novelty claim is not that transformers exist, but that a production-oriented, multi-task DeBERTa pipeline can jointly estimate project strength, skill evidence, and alignment, then expose those estimates through explainable microservices inside a complete hiring product.

1.5 Aim and objectives

Aim: Design, implement, and evaluate an intelligent resume parsing and skill proficiency analysis component that produces structured, explainable, and integration-ready candidate skill profiles for an AI-powered talent acquisition platform.

Specific objectives:

- Define a research problem and requirements model that separates field extraction, proficiency estimation, and explanation.
- Construct and prepare a localized resume skill-context dataset suitable for multi-task supervised learning.
- Implement robust CV text extraction for PDF, DOCX, and TXT inputs with structured candidate field parsing.
- Develop rule-based section and skill detection as a transparent baseline and orchestration layer.
- Train and evaluate a multi-task DeBERTa-v3 model for project strength, skill-evidence strength, and experience–project alignment.
- Integrate Gemini-based explanations that cite CV evidence for proficiency and competency scores.
- Deliver candidate and recruiter-facing skill analysis interfaces and verify them through automated and integration tests.

• Assess research performance using regression and classification metrics, and document commercialization and ethical considerations.

1.6 Research questions or hypotheses

Primary research question:

RQ1: Can a multi-task transformer model that jointly predicts continuous strength scores and ordinal proficiency levels produce accurate and useful skill profiles from parsed resume evidence? Supporting research questions.

RQ2: How does structured skill-context construction (skill, project, experience, certifications) affect scoring quality compared with raw resume text alone?

RQ3: Can hybrid orchestration that combines deterministic parsing with neural scoring improve operational reliability when the model service is unavailable?

RQ4: Do CV-grounded explanations improve the interpretability of proficiency outputs for platform users?

Working hypothesis: A hybrid pipeline that combines alias-aware parsing with a calibrated multi-task DeBERTa model can achieve high held-out accuracy for skill strength estimation while remaining deployable as explainable microservices in a talent acquisition product.

1.7 Scope limitations and assumptions

Scope inclusions:

- Textual resumes in PDF, DOCX, and TXT formats with selectable text.
- Technical skill detection based on a maintained catalogue and alias list.
- Proficiency estimation at skill–project–experience context level using DeBERTa-v3.
- Gemini-based natural-language explanations linked to CV evidence.
- Integration with candidate profile storage and skill-analysis UI. Scope exclusions and limitations:
- Scanned image-only PDFs requiring OCR are out of current production scope.
- The system recommends and explains; it does not autonomously hire or reject candidates.
- Soft-skill inference is lightweight and keyword-assisted rather than deeply psychometric.
- User or stakeholder validation is pending and is left blank in Section 6.3.

- Longitudinal employer outcome validation of proficiency scores is future work.

Assumptions include that uploaded documents are voluntarily provided by authenticated users, that English-dominant technical resumes are the primary corpus, and that human reviewers remain responsible for final hiring judgments.

1.8 Individual contribution

My individual contribution covers the complete research and product lifecycle of the resume intelligence component. This includes problem formulation and gap analysis; dataset preparation and labelling strategy for skill contexts; design of the hybrid parsing and multi-task modelling approach; implementation of cv-extraction service, cv-profile-analysis-service, and resume-strength-model-service; integration with the explanation service; development of the Skill Analysis frontend experience; automated tests; evaluation against held-out metrics; and documentation of 14 commercialization, ethics, and future work. Contributions belonging to teammate components-such as attrition prediction, interview analytics, and job matching models-are acknowledged as dependent interfaces rather than claimed as individual work.

1.9 Thesis structure

Chapter 1 introduces the platform context, individual component, problem, gap, objectives, and scope.

Chapter 2 reviews literature and related systems.

Chapter 3 presents research methodology, data, experiments, metrics, ethics, and reproducibility.

Chapter 4 describes product and solution design.

Chapter 5 reports implementation, integration, testing, and deployment.

Chapter 6 presents results and validation, with Section 6.3 reserved for pending stakeholder validation.

Chapter 7 discusses findings and limitations.

Chapter 8 addresses commercialization and sustainability.

Chapter 9 concludes and outlines future directions.

2 LITERATURE REVIEW AND RELATED WORK

2.1 Review method and scope

The literature review was conducted as a structured narrative survey focused on five themes:

(i) resume and CV information extraction,
(ii) skill ontology and proficiency modelling,
(iii) transformer-based NLP for document understanding,
(iv) explainable AI in human-resource decision support, and
(v) commercial ATS and parsing products.

Sources included IEEE Xplore, ACM Digital Library, Scopus indexed journals, arXiv preprints for recent NLP models, and vendor documentation for product comparison. Priority was given to peer-reviewed work from approximately 2016 onward, with earlier foundational papers retained where they define core concepts such as named entity recognition for resumes [8], [9]. Inclusion criteria required direct relevance to resume parsing, skill extraction, proficiency estimation, or explainable HR analytics. Exclusion criteria removed papers that addressed only generic sentiment analysis, unrelated recruitment chatbots, or pure job-board scraping without document understanding. The goal of the review is not encyclopedic coverage of all NLP, but a focused synthesis that justifies the research gap for this component.

2.2 Domain and theoretical foundations

Recruitment theory distinguishes between person–job fit and person organization fit [10]. Resume screening primarily supports person–job fit by assessing whether claimed competencies align with role requirements. In information systems terms, the resume is an imperfect signal: candidates may understate, overstate, or format evidence inconsistently. Signaling theory therefore warns that presence of a keyword is not equivalent to demonstrated capability [11]. From an NLP perspective, resume understanding combines information extraction, section segmentation, entity linking, and text classification. Classical systems used rules, lexicons, and conditional random fields [8]. Neural systems later applied recurrent networks and transformers to improve generalization across layouts [12], [13]. DeBERTa improves on BERT-style encoders through disentangled attention and enhanced mask decoding, making it attractive for

nuanced evidence scoring tasks [14]. Proficiency modelling can be framed as ordinal classification (Beginner / Intermediate / Advanced) or continuous regression on a bounded score scale. Multi task learning is theoretically attractive because project quality, skill evidence, and experience alignment are related but not identical constructs; sharing an encoder can improve sample efficiency when labels are expensive [15]. Explainable AI literature further argues that decision-support systems in employment contexts should expose reasons and evidence rather than black-box ranks [3], [7], [16].

2.3 Related research

Resume information extraction research has progressed from template-driven parsers to machine-learning extractors capable of identifying names, education, experience, and skills across heterogeneous layouts [5], [9], [17]. Several studies report high field-level F1 for common entities, yet fewer report proficiency estimation beyond skill lists. Skill extraction work often relies on taxonomies such as ESCO or O*NET and focuses on normalization rather than strength scoring [18], [19]. Recent transformer studies apply BERT and variants to resume classification, job recommendation, and candidate ranking [12], [20], [21]. These approaches demonstrate that contextual embeddings outperform bag-of-words baselines on noisy career text. However, many pipelines end at ranking or binary match prediction and do not expose per-skill proficiency with project-level evidence. Explainability research in HR analytics has used SHAP, attention visualization, and natural-language rationales [7], [16], [22]. For resume scoring, a practical form of explanation is evidence quoting: showing which CV spans support a proficiency judgment. Large language models can generate such rationales, but they must be constrained to the candidate document to reduce hallucination risk [23], [24]. In the Sri Lankan and broader South Asian context, digital hiring adoption is increasing, yet published end-to-end open systems that combine localized resume corpora with deployable proficiency models remain limited. This scarcity reinforces the need for a component that is both experimentally evaluated and product integrated. A further strand of related work concerns representation learning for person job fit. Ability-aware neural networks and joint representation models improve ranking quality by embedding candidates and jobs in a shared space [20], [21]. While valuable

for matching, those models typically optimize a match score rather than an inspectable per-skill proficiency profile. This thesis therefore treats matching as a downstream consumer of structured skill outputs rather than as a substitute for proficiency analysis. The distinction matters for fairness and contestability: a recruiter can challenge a skill-level judgment supported by CV evidence more easily than a latent embedding distance. Finally, industrial practice increasingly uses large language models to summarize resumes. Unconstrained summarization can hallucinate credentials or invent skills that are not present in the source document [23], [24]. The proposed component therefore separates deterministic extraction and supervised proficiency scoring from LLM-based explanation, and constrains explanations to the uploaded CV text. This design choice is a direct response to faithfulness risks identified in the literature.

2.4 Existing systems products and techniques

Commercial ATS platforms typically emphasize workflow, compliance, and keyword filters. Parsing add-ons extract contact fields and skill tags, but proficiency is often left to recruiter interpretation or simplistic years-of-experience heuristics [6]. Open-source libraries can extract text from PDF and DOCX reliably for digital documents, yet they do not solve semantic proficiency estimation by themselves.

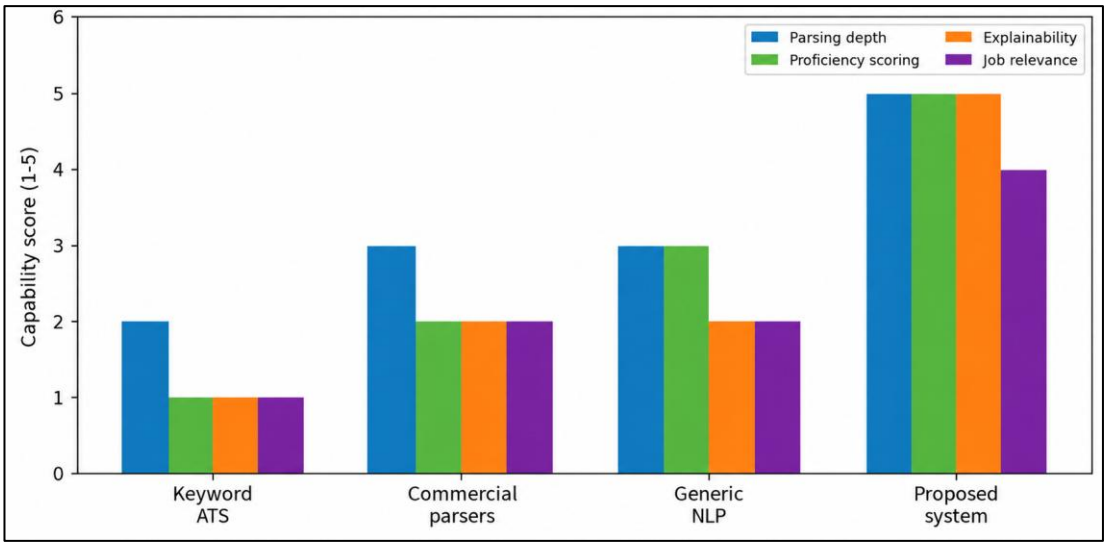


Figure 2: Capability comparison of resume analysis approaches

Table 1:Comparison of existing resume analysis approaches and the proposed component

Approach	Parsing	Proficiency	Explainability	Platform integration	Limitation
Keyword ATS	Low–medium	Mostly absent	Low	High workflow focus	Brittle term matching
Commercial parsers	High fields	Limited / heuristic	Limited	Vendor lock in	Weak evidence scoring
Generic NLP classifiers	Medium	Partial	Model-dependent	Usually standalone	Weak CV grounding
Rule-only extractors	Medium	Heuristic levels	Transparent but shallow	Easy to embed	Miss nuanced evidence
Proposed component	High structured	Multi-task neural + rules	Gemini CV grounded	Microservice platform	OCR and stakeholder study pending

2.5 Gap synthesis and conceptual framework

Synthesizing the review, the conceptual framework for this component has four layers. Layer 1 is document ingestion and text recovery. Layer 2 is deterministic structure recovery: sections, skills, projects, and durations. Layer 3 is multi-task neural proficiency estimation over skill contexts. Layer 4 is explanation and human facing presentation. The framework assumes that neither rules nor neural models alone are sufficient: rules provide reliability and interpretability for extraction, while the transformer provides nuanced scoring over rich contexts. The research gap addressed is therefore the missing bridge between parseable resume structure and trustworthy proficiency analytics inside an operational talent acquisition system. The remainder of this thesis operationalizes that framework through methodology, design, implementation, and evaluation.

3 RESEARCH METHODOLOGY

3.1 Research design

The study follows a design-science and experimental mixed approach. Design science is appropriate because the primary artefact is a working software–ML component that must satisfy utility requirements in a hiring platform . Controlled experiments are used to evaluate the learning artefact on held-out data using predefined metrics. Qualitative product validation with users is planned but not yet completed; accordingly, Section 6.3 remains blank.

The research process comprised:

(1) problem and requirement elicitation from platform goals.

(2) literature-informed architecture selection.

(3) dataset construction and skill-context expansion.

(4) baseline rule-based analysis.

(5) multi-task model training and hyperparameter comparison.

(6) microservice implementation and frontend integration.

(7) automated verification.

(8) quantitative evaluation against test metrics.

3.2 Data participants or study materials

The modelling corpus is a Sri Lankan localized resume dataset prepared for skill-context learning. Metrics artefacts record 4,817 raw resume-linked records expanded into skill-context training rows. After expansion and splitting, the V5 dataset contains 17,146 training rows, 3,704 validation rows, and 3,717 test rows. Each row pairs a target skill with project text, professional experience, certifications, and engineered evidence indicators.

Table 2:Dataset composition for the resume strength model (V5)

Subset	Rows	Role
Raw records	4,817	Source resume-linked materials before context expansion
Training	17,146	Parameter updates and loss optimization

Validation	3,704	Trial selection, early stopping guidance, calibration checks
Test	3,717	Untouched final reporting

Labels cover three related targets: project strength, skill-evidence strength, and experience–project alignment. Each target has a continuous score on a 0–100 scale and an ordinal class in {Beginner, Intermediate, Advanced}. This dual supervision supports both nuanced ranking and human-readable proficiency bands. Software study materials additionally include unit tests with synthetic resumes, API contracts, and persisted MongoDB analysis documents.

3.3 Experimental or study procedure

Model development used microsoft/deberta-v3-base as the shared encoder with a 12-output head: three sigmoid-style regression logits and nine classification logits forming three three-class softmax heads [14]. Inputs were formatted as structured prompts containing skill identity, engineered evidence flags, project text, experience text, and certifications. Sequence length was capped at 384 tokens for latency-aware inference. Three hyperparameter trials (A, B, C) varied encoder learning rate, head learning rate, weight decay, dropout, focal-loss gamma, classification loss weight, and consistency regularization. Trial B was selected using a composite validation selection score that balances MAE, accuracy, balanced accuracy, macro-F1, and quadratic weighted kappa. The selected checkpoint was then evaluated once on the held-out test partition. System experiments exercised the full pipeline: upload a resume, extract text, run rule-based analysis, enrich with model scores, request explanations, persist results, and render the skill analysis page. Failure modes were tested by disabling the model service and confirming that rule-based outputs still return useful profiles with degraded model status flags.

3.4 Measures and evaluation metrics

Table 3:Evaluation metrics used in the research

Metric	Target type	Interpretation
MAE / RMSE / R^2	Regression	Continuous score fidelity on 0-100 scale
Within ±5 / ±10	Regression	Operational tolerance for recruiter-facing scores
Accuracy	Classification	Overall class correctness
Balanced accuracy	Classification	Robustness under class imbalance
Macro-F1	Classification	Per-class harmonic mean averaged equally
Quadratic Weighted Kappa	Ordinal classes	Agreement emphasizing ordinal distance
Technical competency score	Product metric	Aggregated profile strength for UI
Service availability flags	Product metric	Live / unavailable / not-run statuses

Classification metrics are reported separately for Project Strength, Skill Strength, and Experience–Project Alignment. This separation is essential because class balance differs across heads; for example, Advanced project-strength labels are rare in the test set, which lowers balanced accuracy even when overall accuracy remains high.

3.5 Data analysis

Quantitative analysis used the exported metrics.json and best_validation_metrics.json artefacts produced during training, together with hyperparameter_comparison.csv. Confusion matrices and per-class precision/recall were inspected to identify minority-class weaknesses. Product analysis inspected sample pipeline outputs for coherence between detected skills, model levels, and explanation text. No claim of causal hiring improvement is made from offline metrics alone.

3.6 Ethics privacy security and responsible AI

Resumes contain personal data, including names, contacts, education, and employment history. The component therefore assumes authenticated access, purpose limitation to recruitment support, and storage minimization through profile scoped persistence. Model outputs are decision-support artefacts, not automated adverse actions. Explanations are constrained to the candidate CV to reduce unsupported claims. Bias risks remain because training data and skill catalogues can under-represent non-software occupations or non-English evidence; these risks are disclosed as limitations rather than claimed to be solved [4], [16]. Security controls include service isolation, request identifiers for traceability, input validation, and rejection of unsupported or empty documents. Secrets for explanation-model access are handled through environment configuration rather than source code. Responsible AI practice in this component emphasizes contestability: users can inspect evidence and status flags when model enrichment is unavailable.

3.7 Reproducibility and research artefacts

Reproducibility artefacts include the trained model directory (tokenizer, config, weights, metrics), training notebooks under the Models workspace, microservice source code, unit tests, and exported CSV summaries such as per_skill_summary.csv and test_predictions.csv. Runtime reproducibility is supported by containerized services and explicit model version tags (resume_strength_model_v5). Randomness in deep learning training implies that exact bit-level replication may vary across hardware, but reported metrics correspond to the checked-in evaluation artefacts used by the deployed service.

4 PRODUCT AND SOLUTION DESIGN

4.1 Requirements

- Upload and extract text from PDF, DOCX, and TXT resumes.
- Parse candidate fields and major CV sections.
- Detect technical skills using alias-aware catalogue matching.
- Estimate skill proficiency and related strength scores via ML service.
- Compute an aggregate technical competency score.
- Generate CV-grounded explanations for scores.
- Persist analysis results to the candidate profile.
- Display skill analysis to candidates and support admin review views.
- Expose clear service status when model or explanation is unavailable.
- Provide structured outputs usable by matching and other modules.
- Usability: Present scores and evidence without requiring ML expertise.
- Reliability: Degrade gracefully if ML/explanation services fail.
- Performance: Support interactive analysis within practical web latencies.
- Security/Privacy: Ensure authenticated access and purpose-limited processing.
- Maintainability: Use microservice boundaries and explicit versioning.
- Fairness Transparency: Expose evidence and limitations to reviewers.

4.2 Component architecture within the complete solution

Figure 3 illustrates the architecture of the implemented Intelligent Resume Parsing and Skill Proficiency Analysis component within the Talent AI system. The component processes uploaded resumes through document extraction, NLP-based entity recognition, skill normalization, and proficiency analysis. The resulting structured candidate profiles are made available to other recruitment modules.

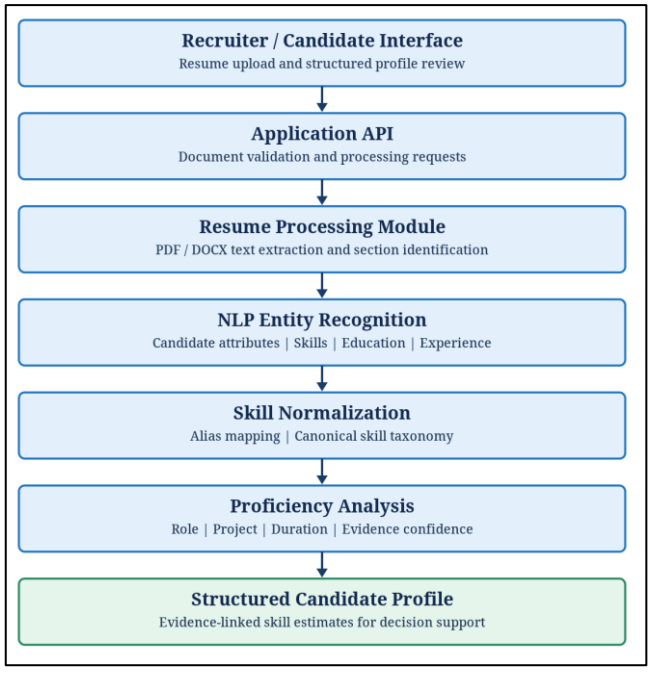


Figure 3:Resume-to-profile architecture

The architecture separates the user interface, application API, resume processing, NLP analysis, and profile generation into distinct functional stages. This modular structure improves maintainability and allows each processing stage to be tested independently.

Resume documents are received through the application interface and passed to the processing module. Extracted information is analyzed to identify candidate attributes and supporting skill evidence. The generated profile provides standardized candidate information for integration with job matching, interview evaluation, and hiring attrition analysis components.

The component handles uploaded resume content as untrusted input. Document validation, error handling, and structured output management support reliable processing. Skill proficiency results are presented as evidence-based estimates rather than verified qualifications or final hiring decisions.

4.3 Design alternatives and decisions

The implemented Intelligent Resume Parsing and Skill Proficiency Analysis component consists of document extraction, NLP-based entity recognition, skill normalization, proficiency analysis, and candidate profile generation. These modules work together to convert unstructured resume content into organized candidate information.

Resume text extraction is an essential stage of the system, as PDF documents may contain different layouts, fonts, and formatting structures. Libraries such as pdfplumber and PyMuPDF provide approaches for retrieving textual information. Extraction quality is important because missing or incorrectly ordered text can affect subsequent NLP processing.

Dictionary-based matching provides a straightforward method for identifying known skills, while Conditional Random Fields (CRF) support entity recognition through token features and sequence relationships. Transformer-based models offer an alternative for capturing broader linguistic context but generally require greater computational resources. These approaches involve trade-offs in accuracy, complexity, and processing efficiency.

MongoDB provides a flexible document-based structure suitable for storing candidate attributes, extracted skills, work experience, and proficiency-related evidence. Compared with a fixed relational schema, document storage can accommodate differences between candidate profiles. Data validation, indexing, and access control remain important for reliable information management.

4.4 Detailed design

- Extraction and parsing design

Extraction accepts multipart file uploads and routes by extension. PDF text is recovered through pdf-parse; DOCX through mammoth; TXT through UTF-8 decoding. Empty or non-selectable documents are rejected with actionable errors. Candidate fields such as name, contacts, and highlighted content are derived from normalized text. Profile analysis then segments education, experience, projects, certifications, and skills.

- Skill catalogue and evidence heuristics

Skills are matched using a canonical catalogue with aliases (for example, React, ReactJS, React.js). For each detected skill, the parser estimates duration from nearby year/month phrases, counts occurrences, checks project usage, and assigns a provisional rule-based level and confidence. Job-aware relevance tags (must-have, nice-to-have, additional) are attached when a job object is supplied.

- Multi-task scoring design

For each skill, the orchestrator builds one or more contexts pairing the skill with related projects and experience. The Python service formats structured model input, runs DeBERTa inference, applies sigmoid scoring for continuous heads, and softmax classification for ordinal heads. Results are merged back into skill and project objects, replacing or enriching rule-based levels when model scores are available.

- Explanation and presentation design

The explanation client posts raw CV text and analysis JSON to the explanation service. Returned rationales are attached under an explainability object with live/unavailable status. The frontend Skill Analysis page renders competency gauges, level

distributions, skill bars, model output review panels, and expandable “Why this score?” evidence.

4.5 Standards and professional practice

Development followed modular service boundaries, explicit validation of inputs, health endpoints, and testable pure functions for parsing logic. API responses use consistent success envelopes and request identifiers. From a professional ethics standpoint, the design keeps humans in control, discloses model status, and avoids presenting speculative claims as facts. Documentation and version tags support maintainability for subsequent teams.

5 IMPLEMENTATION INTEGRATION AND TESTING

5.1 Development environment and technology stack

Table 4:Technology stack for the individual component

Layer	Technologies
Frontend	React, Vite, React Router, component CSS modules
Orchestration services	Node.js, Express-style services, MongoDB
Extraction libraries	pdf-parse, mammoth
ML service	Python, FastAPI, PyTorch, Hugging Face Transformers
Model	microsoft/deberta-v3-base fine-tuned multi-task checkpoint V5
Explainability	Gemini via resume-explanation-model-service
Quality tools	Node test runner assertions, service health checks
Ops	Containerized local/service deployment, environment configuration

5.2 Component implementation

Implementation is distributed across four primary code areas. The cv extraction-service implements extractRawText and extractCv. The cv-profile analysis-service implements analyzeCv for deterministic analysis and analyzeCvWithModel for neural enrichment plus explanation. The resume-strength model-service exposes /health, /predict, and /predict/batch endpoints backed by ResumeStrengthRunner. The frontend implements ResumeStrengthSummary, and ModelOutputReview. SkillAnalysisPage, A notable implementation detail is graceful degradation. If no skills are detected, the model is marked not-run. If inference fails, the base analysis remains available with model status unavailable. If Gemini explanation fails, saved model scores remain visible. This design keeps the product usable under partial outages. Implementation specifics aligned to the source services include: (i) multipart upload validation and extension-based routing in the extraction service; (ii) catalogue-driven skill detection with alias normalization and duration heuristics in the profile-analysis service; (iii) batch-capable FastAPI inference with health probes exposing modelLoaded and

degraded states; and (iv) React views that surface 30 competency gauges, level distributions, ModelOutputReview panels, and expandable evidence rationales. Persistence uses MongoDB documents keyed to the candidate profile so that analyses can be reopened without recomputation and re-triggered when model or explanation fields are incomplete after an upgrade. The hybrid analysis method identifier recorded in enriched outputs (deberta v3-base-multitask-with-rule-based-parsing) and the model version tag (resume_strength_model_v5) provide auditability for later comparison of checkpoints. Unit tests under the profile-analysis service verify deterministic construction of explainable skills and non-zero competency scoring on synthetic resumes containing Python, React, and Docker evidenc

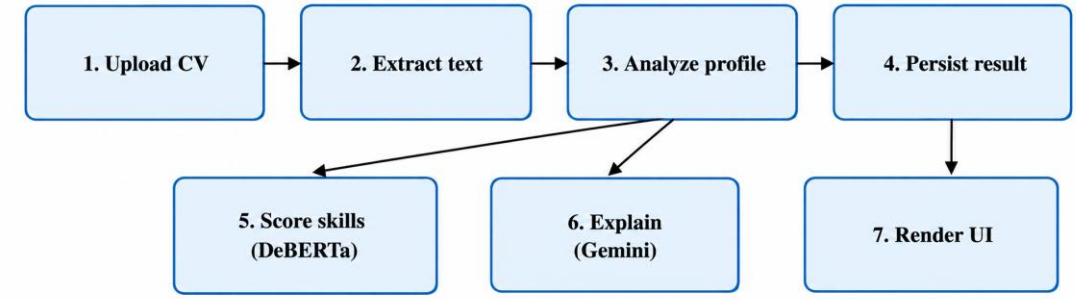


Figure 4:End-to-end runtime flow for resume parsing and skill analysis

5.3 Integration with the other components

Structured candidate profiles are the interface contract with matching and other recruitment components. Downstream services should receive normalized skills and traceable evidence, not raw unreviewed personal information by default. A profile version identifier prevents silent inconsistencies when the parser is updated. Figure 5.1 shows the intended exchange.

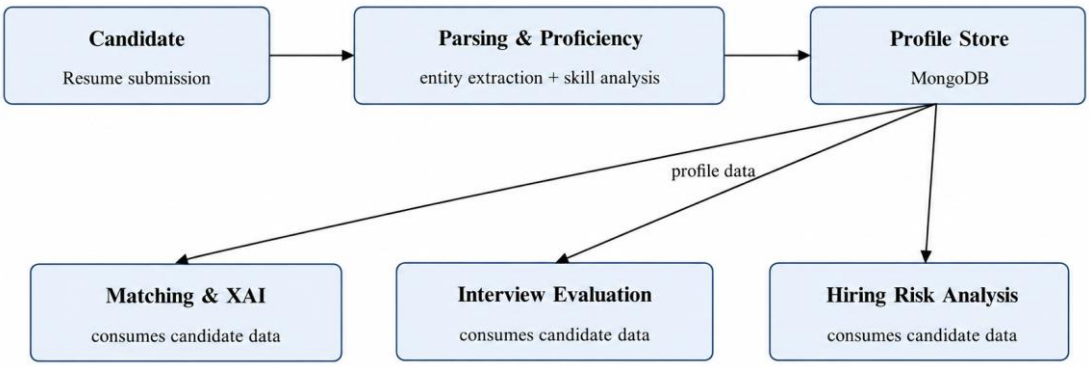


Figure 5:Component integration and data flow

The shared candidate profile is the integration boundary. Downstream matching should receive canonical skill identifiers and provenanceaware estimates, while interview and risk components may consume other validated profile fields. A contract test should verify required keys, data types, optional fields and schema version. 37 Consumers must not assume that a missing skill means the candidate is unqualified. Integration success should be demonstrated with end-to-end traces and reproducible test inputs. Versioning and failure isolation. An updated normalization taxonomy or proficiency model can change profile outputs even if the API shape remains stable. Version fields should therefore identify the schema and model or rule set used. Downstream services should degrade gracefully if proficiency is unavailable, for example by displaying extracted skills without inferred levels. This design avoids turning a failure in one optional analysis stage into a complete loss of candidate information.

5.4 Verification and software testing

Test ID	Scenario	Expected result
TC01	Analyze synthetic CV with Python/React/Docker	Skills detected; competency > 0
TC02	Mark must-have skill present in CV	Relevance tagged Must-have
TC03	Upload unsupported extension	415/validation error
TC04	Upload empty/scanned PDF text	422 actionable error
TC05	Model service down	Base analysis returned; model unavailable
TC06	Explanation service down	Scores visible; explainability unavailable
TC07	No skills detected	Model not-run reason recorded

TC08	Skill analysis page without CV	Prompt user to upload CV
TC09	Admin opens skill analysis route	Redirected away from candidate page
TC10	Batch predict with empty items	400-level validation failure

Automated unit tests validate deterministic analyzeCv behaviour, including name recovery, project extraction, skill relevance tagging, and non-zero competency scoring. Manual exploratory testing verified UI states for loading, error, empty profile, and enriched model output.

5.5 Deployment and operation

Operationally, each service can be started independently with environment variables for model directory, device (CPU/CUDA), maximum sequence length, Mongo connection, and explanation service URL. Health endpoints expose modelLoaded and degraded states for monitoring. The model directory includes tokenizer files, configuration, metrics, and calibration parameters so that deployment does not depend on rebuilding training notebooks. Logs and request IDs support incident diagnosis during integration.

6 RESULTS AND VALIDATION

6.1 Research results

Hyperparameter comparison identified Trial B as the selected configuration with validation selection score 0.8943, marginally above Trial A (0.8938) and clearly above Trial C (0.8861). Trial B used encoder learning rate 6e-6, head learning rate 3.5e-5, dropout 0.15, classification lambda 1.3, and focal gamma 2.25. Although Trial A reported lower MAE, Trial B offered stronger balanced accuracy and macro F1 trade-offs under the project selection criterion.

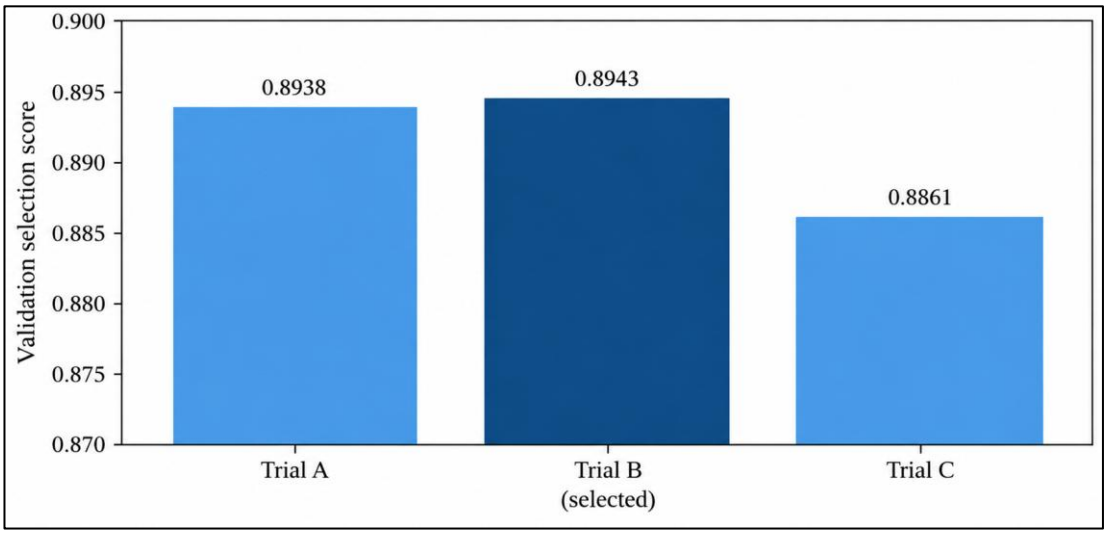


Figure 6:Hyperparameter trial comparison for model V5

Table 5:Hyperparameter search trials and validation selection scores

Trial	Enc. LR	Head LR	Dropout	Val MAE	Val MAE	Val Macro F1	Selection
A	8e-6	4e-5	0.10	2.16	0.977	0.868	0.8938
B (selected)	6e-6	3.5e-5	0.15	2.67	0.954	0.874	0.8943
C	5e-6	3e-5	0.10	2.66	0.961	0.865	0.8861

On the held-out test set, regression performance was strong across all three continuous heads. Skill-evidence strength achieved MAE 2.50, RMSE 2.99, and R^2 33 0.983.

Project strength achieved MAE 2.23 and R^2 0.899. Alignment was harder (MAE 3.47) but still highly correlated (R^2 0.967). Average within-±10 tolerance reached 99.15%, indicating that most score errors remain within a recruiter interpretable band.

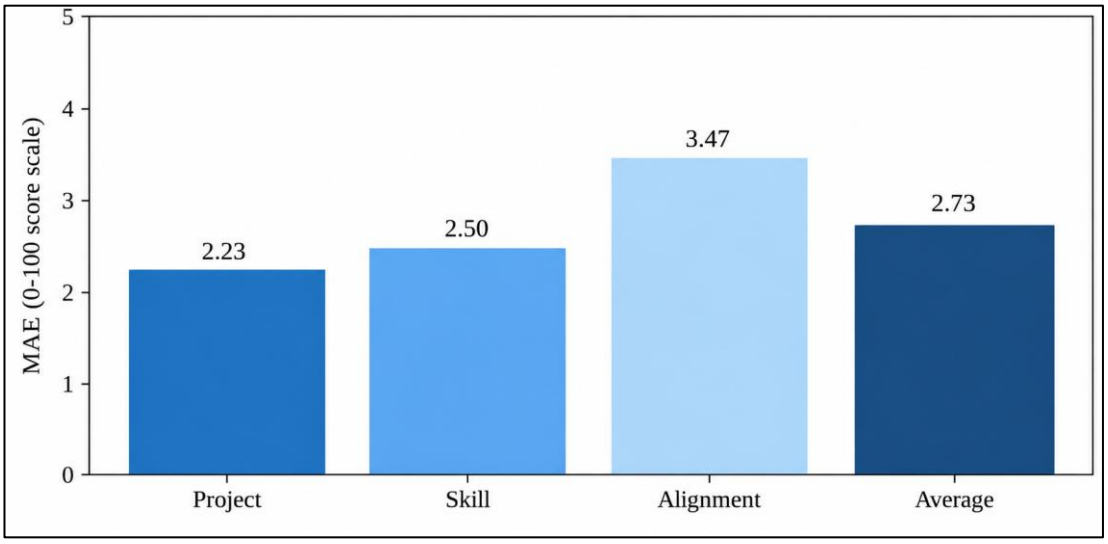


Figure 7:Test-set regression MAE by task head

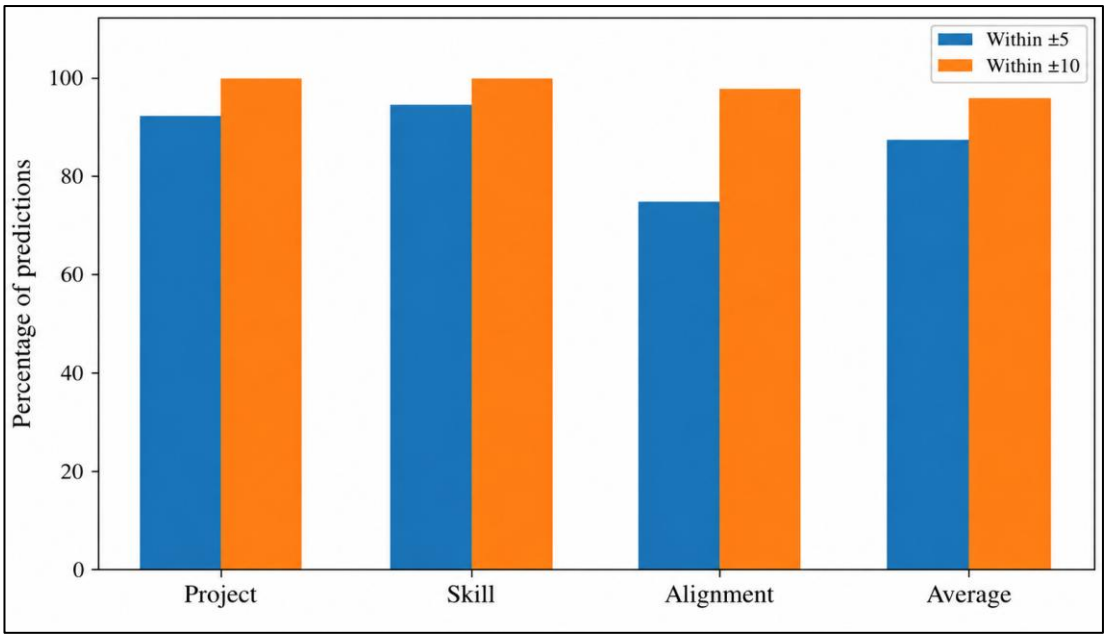


Figure 8:Regression tolerance accuracy on the test set

Task	MAE	RMSE	R^2	Within ±5	Within ±10
Project strength	2.23	2.88	0.899	92.12%	99.60%
Skill evidence	2.50	2.99	0.983	94.32%	99.60%

Alignment	3.47	4.25	0.967	75.38%	98.25%
Average	2.73	-	-	87.27%	99.15%

Classification results show especially strong skill-strength recognition (accuracy 98.98%, balanced accuracy 98.03%, macro-F1 0.982). Alignment classification also performed well (accuracy 94.92%, macro-F1 0.939). Project strength classification accuracy remained high (90.34%), but balanced accuracy dropped to 68.71% because Advanced project labels are rare (support 20 in the test split). This imbalance is an important validity caveat and motivates future resampling or cost-sensitive refinements.

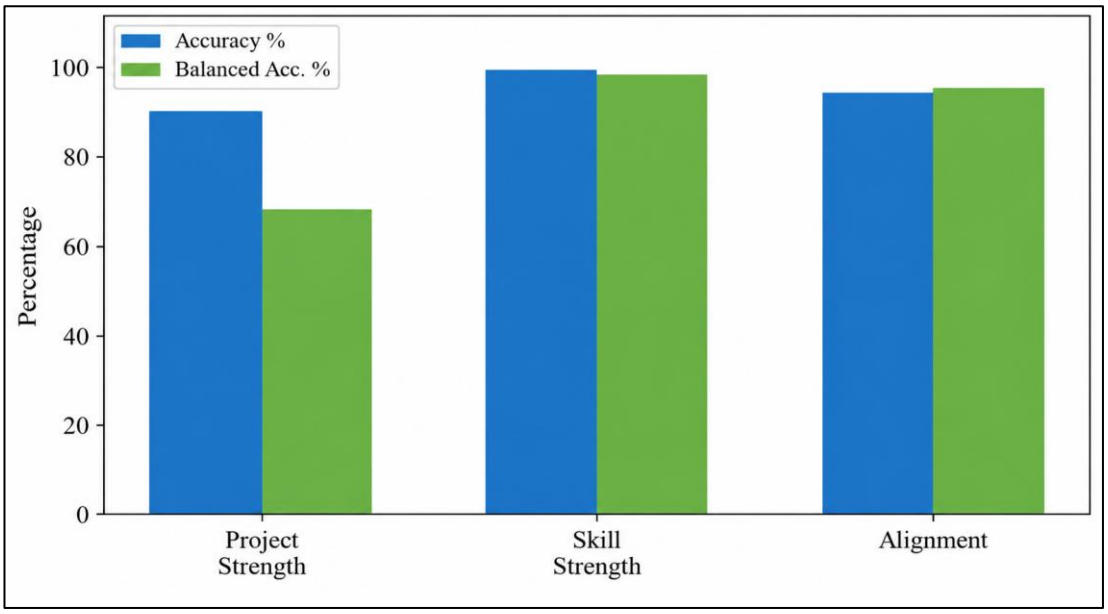


Figure 9:Test-set classification performance

Table 6:Held-out test classification metrics

Task	Accuracy	Balanced Acc	Macro-F1	QWK
Project Strength	90.34%	68.71%	0.652	0.820
Skill Strength	98.98%	98.03%	0.982	0.993
Experience Project Alignment	94.92%	95.42%	0.939	0.949

6.2 Product and technical results

Product implementation results confirm that the component meets its core functional requirements in the integrated platform. Candidates can upload resumes, obtain structured skill profiles, and open a dedicated Skill Analysis view showing competency gauges, skill-level distributions, evidence bars, and explanation panels. Admin/application views can inspect candidate skills using the same analysis objects. Technical status badges communicate whether model enrichment and Gemini explanations are live. Deterministic tests passed for explainable skill construction and competency scoring on synthetic resumes. Integration behaviour under partial failures matched the design: the UI remains informative when explanation generation is temporarily unavailable. The analysis method string recorded in enriched outputs (deberta-v3 base-multitask-with-rule-based-parsing) provides an auditable marker of the hybrid approach.

6.3 User or stakeholder validation

Not yet completed

6.4 Integration results

Integration testing with the wider platform demonstrated that resume intelligence outputs are consumable beyond the skill-analysis page. Profile persistence enables re-opening analyses without recomputation, while re-analysis can be triggered when explanations or model fields are missing. The component therefore behaves as a reusable service boundary rather than a page-local script. No blocking interface defects were observed in the core upload–analyze–render path during development verification.

6.5 Summary of findings

Objective theme	Evidence summary	Status
Dataset + multi-task model	V5 metrics on 3,717 test rows; strong skill metrics	Achieved
Hybrid parsing pipeline	Extraction + catalogue + orchestration implemented	Achieved

Explainability	Gemini CV-grounded explanations integrated with fallback	Achieved
Product UI + APIs	Skill analysis page and services operational	Achieved
Stakeholder validation	Reserved for later completion	Pending (blank)

Overall, the research results support the hypothesis that hybrid multi-task modelling can estimate skill proficiency with high held-out fidelity, while the product results show that those estimates can be operationalized inside a talent acquisition platform with explicit uncertainty and explanation channels.

7 DISCUSSION

7.1 Interpretation of findings

The strongest signal in the results is the skill-strength head, where both regression and classification metrics are excellent. This suggests that the structured skill-context representation—aliases, mention counts, project linkage, certifications, and experience—provides discriminative evidence for proficiency. Alignment remains slightly harder, which is expected because consistency between experience and projects is semantically subtle and can be sparsely expressed in resumes. The project-strength minority-class issue indicates that overall accuracy can conceal weakness on rare Advanced projects. In deployment, this means Advanced project badges should be interpreted cautiously and always in conjunction with continuous scores and evidence snippets. The hybrid architecture mitigates total failure risk: when neural enrichment is unavailable, rule-based levels still provide a transparent baseline.

7.2 Comparison with literature baselines and existing solutions

Relative to keyword ATS behaviour described in literature and industry practice, the component advances from binary skill presence to graded proficiency with evidence [1], [6]. Relative to generic resume classifiers, it preserves per-skill interpretability rather than collapsing a CV into a single rank [20], [21]. Relative to rule-only systems, it captures nuanced cues such as ownership language, complexity markers, and cross-section consistency through learned representations [14], [15]. The explanation layer aligns with responsible AI recommendations for employment technologies by exposing reasons and CV evidence rather than only numeric outputs [3], [16]. Unlike purely post-hoc feature-attribution dashboards, the chosen design optimizes for recruiter-readable narratives, accepting the trade-off that LLM explanation quality must be monitored for faithfulness.

7.3 Research contribution

The research contribution is a validated hybrid method for resume skill proficiency analysis that combines deterministic parsing with a 12-output multi-task DeBERTa model trained on localized skill contexts. The work contributes empirical evidence—MAE, tolerance accuracy, macro-F1, and QWK—across project, skill, and alignment

heads, and documents minority-class limitations with confusion matrix awareness rather than reporting only headline accuracy.

7.4 Product contribution

The product contribution is a deployable microservice suite and user experience that transform uploaded resumes into explainable skill profiles inside an integrated talent acquisition system. Concrete deliverables include extraction and analysis services, model serving, explanation integration, persistence, and the Skill Analysis interface. These artefacts make the research reusable by sibling components and future maintainers.

7.5 Limitations and threats to validity

- External validity: results are tied to the localized dataset and English dominant technical resumes.
- Construct validity: proficiency labels are dataset-defined approximations of real workplace skill.
- Class imbalance: Advanced project-strength support is thin, weakening balanced metrics.
- OCR gap: image-only scanned CVs are not handled in the current extractor.
- Explanation faithfulness: Gemini rationales require ongoing audit against CV evidence.
- Stakeholder validity: user validation is pending and currently blank in Section 6.3.
- Outcome validity: no longitudinal proof yet that scores improve hiring quality or fairness.

7.6 Implications

For practitioners, the findings imply that resume intelligence should expose uncertainty, evidence, and service health rather than a single opaque rank. For researchers, the results encourage multi-task formulations that separate related proficiency constructs. For policymakers and HR leaders, the component illustrates a practical path toward more contestable AI assistance in hiring, provided human oversight and bias monitoring remain mandatory.

8 COMMERCIALISATION AND SUSTAINABILITY

8.1 Value proposition and target users

8.2 Market and competitive position

8.3 Business or sustainability model

8.4 Scalability intellectual property and compliance

8.5 Contribution to the Sustainable Development Goals

9 CONCLUSION AND FUTURE DIRECTIONS

9.1 Conclusion

This thesis presented the design, implementation, and evaluation of an Intelligent Resume Parsing and Skill Proficiency Analysis component for an AI Powered Intelligent Talent Acquisition System. By combining robust document extraction, alias-aware skill parsing, multi-task DeBERTa scoring, and CV-grounded explanations, the component converts heterogeneous resumes into structured profiles that humans and downstream services can use. Held-out experiments demonstrate strong predictive fidelity, especially for skill-evidence estimation, while product engineering results confirm operational integration with graceful degradation.

9.2 Achievement of objectives

Table 7:Achievement of research and product objectives

Objective	Achievement
Problem/requirements definition	Fully achieved
Localized dataset preparation	Fully achieved
CV extraction for PDF/DOCX/TXT	Fully achieved (OCR excluded)
Rule-based section/skill analysis	Fully achieved
DeBERTa multi-task modelling	Fully achieved with reported metrics
Gemini explanation integration	Fully achieved with fallback behaviour
UI + platform integration + tests	Fully achieved for core flows
Stakeholder validation	Not yet completed (Section 6.3 blank)

9.3 Individual research and product contributions

Individually, this work contributes a complete research-to-product pathway for resume intelligence: from gap analysis and dataset-centred experimentation to microservices, UI, and evaluation. The distinguishing research product is the V5 multi-task proficiency model embedded in a hybrid, explainable parsing pipeline rather than a standalone notebook result.

9.4 Future directions

Future improvements to the implemented Intelligent Resume Parsing & Skill Proficiency Analysis component can focus on increasing extraction accuracy, improving contextual skill recognition, and strengthening proficiency estimation. Further research can compare CRF-based and contextual NLP approaches using independently annotated datasets to evaluate their effectiveness across different resume formats and professional backgrounds.

Additional enhancements may include support for scanned and multilingual resumes, improved skill taxonomy management, and confidence-based proficiency estimation. Validating estimated proficiency levels against independent practical assessments could also improve the reliability of skill analysis.

Future studies can involve recruiter usability testing, privacy and fairness assessments, and evaluation of how supporting evidence influences recruitment decisions. These improvements should be guided by identified system limitations and evaluation findings to support the component's continued development and practical adoption.

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GLOSSARY

Alias-aware skill matching - Detection of a canonical skill using multiple surface forms (for example, Node.js / NodeJS).

Graceful degradation - Returning useful partial results when a dependent ML or explanation service is unavailable.

Skill context - A structured model input pairing one skill with project, experience, and certification evidence.

Technical competency score - Aggregate 0–100 indicator summarizing evidence based technical strength of a CV profile.

Multi-task head - Shared-encoder architecture that predicts several related outputs in one forward pass.

APPENDICES

Appendix A Research instruments or extended results

This appendix summarizes the extended research instruments and quantitative artefacts associated with the deployed V5 resume-strength checkpoint. The complete machine-readable outputs, including test_predictions.csv, per_skill_summary.csv, and hyperparameter_comparison.csv, are retained with the model directory so that the reported Chapter 6 metrics can be independently inspected and reproduced.

Across the three supervised task heads, the extended classification review shows a clear performance pattern. Skill Strength remains the strongest operational head, with near-ceiling accuracy and macro-F1 on the held-out test partition, indicating that the structured skill-context representation is highly discriminative for proficiency estimation. Experience–Project Alignment also achieves high accuracy, although its continuous-score MAE is slightly higher than the other heads, which is consistent with the subtler semantic nature of cross-section consistency in resumes. Project Strength retains high overall accuracy, but Advanced-class support is thin in the test split, with only twenty Advanced project labels available; this imbalance reduces balanced accuracy and is therefore treated as an explicit validity caveat rather than as a hidden failure. Probability-calibration factors for the Advanced classes are retained in metrics.json to support transparent post-training inspection.

The parser relies on an alias-aware skill catalogue that maps surface variants onto canonical skill names before scoring. Representative examples include JavaScript mapped from javascript and js; TypeScript from typescript and ts; React from react and reactjs; Node.js from node.js, nodejs, and node js; PostgreSQL from postgresql and postgres; Machine Learning from machine learning and ml; and CI/CD from ci/cd and continuous integration. This normalization reduces duplicate skill entries caused by spelling and branding variation and improves interoperability with downstream matching services.

Appendix B Academic integrity and AI use disclosure

In accordance with module policy, generative AI tools were used as assistive instruments for drafting support, language refinement, and document structuring of this thesis. All research claims, metrics, architecture descriptions, and implementation details were verified against the actual project source code, model artefacts (metrics.json, hyperparameter_comparison.csv), and experimental outputs. The candidate remains solely responsible for the accuracy, originality, citations, analyses, and conclusions presented in this thesis.

Tools and purposes:

- Generative AI assistants: drafting support, restructuring text to the official thesis template, and language polishing.
- Python document libraries: formatting DOCX output to SLIIT margin/font conventions and generating figures.
- Project source code and model metric files: ground truth for technical and experimental statements.

Verification performed: cross-check of service names, endpoints, model version (V5), dataset split counts, and reported MAE/accuracy/F1/QWK values against checked-in artefacts; alignment of chapter structure with the Individual Thesis Content template; reservation of Section 6.3 as blank pending stakeholder validation